Practical No. 10

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Batch: B2

Roll no:33

Aim:version control using Git& Github

What is difference between Git&Github

Ans : Git vs GitHub

Git:

Git is a Version Control System (VCS)

- Created by Linus Torvalds in 2005
- Works completely offline on your local machine
- Tracks every change made to your code
- Stores full project history locally
- Free and open-source
- Used via command line (terminal)
- Alternatives: Mercurial, SVN

GitHub:

GitHub is a cloud-based hosting platform for Git repos

- Founded in 2008, bought by Microsoft in 2018
- Requires internet connection
- Adds collaboration features: Pull Requests, Issues, Actions
- Stores your code remotely on servers
- Has free and paid plans
- Alternatives: GitLab, Bitbucket

2. Explain the difference between:

- git add

- git commit

Ans : git add:

• Moves filesfromWorking Directory → Staging Area

• Tells Git whichchanges to include in next commit

• Does NOT savepermanently

• Can stage onefile,multiple files, or all files

• Reversible usinggitrestore –staged

Commands:

Git add file.txt → one file

Git add . → all files

Git add *.js → all JS files

Git add -p → select parts interactively

•**g** M**it** o**cvoem**s **m**fil**iet:**s from Staging Area → Local Repository
• Creates a permanent snapshot in history
• Each commit gets a unique SHA-1 hash ID

• Requires a commit message

• Records author, timestamp, and changes Commands:

Git commit -m "message" → basic commit

Git commit –amend → edit last commit

Git commit am "message" → add +commit together

3.What is the staging area in Git?

Ans :

• Also called theIndex

• It is a middle layer between working directory and repository

• Physically stored in .git/index file

• Lets you selectively choose what to commit

• You can modify 10 files but commit only 2

• Helps create clean, logical commits

• You can review staged changes before saving

Commands:

Git status → see staged/unstaged files

Git diff –staged → see staged changes

Git restore –staged file → unstage a file -

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• Modified → tracked file with new changes

• Staged → ready to be committed

• Committed → permanently saved in history

4**. What happens if you modify a file after committing it?**

Ans : The old committed version stays safe in history

• Modified file appears as "modified" in working directory

• Git detects change by comparing file hash with last commit

• Git status shows it under "Changes not staged for commit"

• You must add and commit again to save new version

• Each commit gets a new unique hash

• Old commit is still accessible anytime

• You can go back using git checkout <old-hash>

• Nothing in Git is truly lost after committing

• To fix last commit quickly: git commit –amend

Process after modifying:
Modify file → git add → git commit → new snapshot created

5.Difference between:

- git push

- git pull

Ans : git push:

• Uploads localcommits→ remote repository (GitHub)

• Shares yourworkwithteammates

• Only pushescommittedchanges (not just staged)

• May be rejectedifremote has newer commits

Commands:

Git push origin main → push to main branch
Git push origin branch-name → push specific branch
Git push -u origin main → set default upstream
Git push –force → force overwrite (dangerous)

Git pull:

• Downloads remote changes → local machine

• Keeps your local repo up to date

• Combines: git fetch + git merge

• Can cause merge conflicts if same lines were edited Commands:

Git pull origin main → pull from main

Git pull –rebase → pull and rebase instead of merge

6 .What is Version Control? Differentiate between Centralized v/s Distributed VCS.

Ans :

Version Control:

• System that records changes to files over time

• Tracks who, what, when, and why something changed

• Allows reverting to any previous version

• Enables parallel work without overwriting each other

• Essential for team software development

Centralized VCS (CVCS):

• Onecentral server holdsthe entire repository

• Developers check out only one version (not full history)

• All operations need server/internet connection

• Examples: SVN, CVS, Perforce

Advantages:

• Simple to understand

• Admin has full control

• Good for binary/design files

Distributed VCS (DVCS):

• Developer has full copy of entire repo including history

• Most operations happen locally (fast) • Remote server used only

for syncing/sharing

• Examples: Git, Mercurial

Advantages:

• Fully offline capability

• Every clone = complete backup

• Very fast operations

• Easy branching and merging

• No single point of failure

Theory -

Step by Step Procedure

1. Installation of Git
2. Configure username & email
3. Create local repository
4. Create files
5. Add files to staging
6. Commit changes
7. Create GitHub repository
8. Link local repo to GitHub
9. Push files

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